

**DOES CCI HAVE AN IMPACT ON ECONOMIC GROWTH? THE OECD
EXPERIENCE FOR THE YEAR 2019**

KÜBRA ÇOBAN, NECLA NUR ÜNLEYEN, YUNUS EMRE SARIDUMAN

Department of Economics, Marmara University

29.01.2021

1. Introduction

There is an ongoing issue that questions whether consumer confidence index (CCI) has an impact on economic growth or not. The main purpose of this analysis is to detect the relationship between CCI and economic growth in the OECD countries. There are many studies investigating the determinants of economic growth. Most of the time the explanatory variables include inflation rate, interest rate, exchange rate, and unemployment rate. Together, they have significant influence on GDP growth rate.¹

We mainly build our model on explanatory variables as consumer confidence index (CCI), foreign direct investment (FDI), unemployment rate and short-term interest rate. As traditional way did, economic growth is represented by the annual GDP growth rate in this paper. The base year is 2019 since the current data for 2020 is not available yet. Consumer confidence index is a significant index for evaluation of economic growth. Consumer's expectations about the future and even today may have influences on economic situations. Just as principles of economics, law of demand, states that when demand increases, output level increases. Thus, the bigger CCI level leads to more expenditures and thus contributes to economic growth positively.

In this study we discuss as CCI grows in a country then it contributes economic growth positively in OECD countries. In the general sense, OECD members are high-income countries with leading scores in Human Development Index (HDI). Also, CCI levels are relatively high in advanced countries. So, we choose OECD countries to address questions about the effect of CCI on the economy. Many of the OECD countries have CCI data but in Nordic countries; Canada, Iceland and Norway, the current data is not available. Canada has another index for consumer confidence entirely different from other countries, this is why it is not in the regression.

We were expecting to find a linear relationship between CCI and GDP growth (economic growth).

The main findings of the paper are as follows: GDP growth has not a normal distribution but bimodal distribution among OECD countries due to the countries with highest GDP growth rate in 2019. When we examine the CCI and GDP growth relationship, it appears that

¹ Arrazy, M. R. (2013). Factors That Influence GDP Growth Rate in Six Asia's Developing Countries From 2007 To 2011 (Doctoral dissertation, President University).

there is a strong link between them as highly similar to linear relationship. Other variables, FDI, short term interest rate and unemployment rate have no statistically significant effect on economic growth. In other words, we can conclude that CCI and economic growth are linked strongly, high CCI levels escalates GDP growth according to the empirical study. Our main question is supported by the empirical results as it was expected.

The rest of the paper proceeds as follows. Section 2 reviews literature review on the relationship between growth and CCI mainly, also other determinants of growth such as FDI, short term interest rate and unemployment rate. We discuss the data in Section 3 and Section 4 describes a framework for our modelling and the econometric methods used in this study. We present the main results in Section 5. Finally, Section 6 summarizes conclusions.

2. Literature Review

In literature, the link between economic growth and CCI is significant. In the study of EU countries, when consumers have more confidence on initial and future expectations about the economic situation, their demand increases and the demand rises economic growth. Consumer confidence and economic growth positively related. (Islam & Mumtaz, 2011)

Another study that examines the US data with more than 50 years (Mazurek, & Mielcová, 2017), found similarly with other studies, that changes in consumer confidence affects the consumer's demand and spending, affects *GDP with a lag of 3 or 6 months*. However, the authors are suspicious about the short-term influence of CCI on GDP. An article explains the relationship between consumer confidence index and GDP growth in the period of December 2002-2010 using the VAR method. The study is centered on the less developed country which is Trinidad and Tobago. According to data, there was a strong relationship between GDP growth and consumer confidence index in the short run. (Kelvin A Sergeant, 2011)

Economic growth is a very complex topic itself in an open economy, FDI comes to the forefront at this point. Especially in developing countries, heterogeneity is the characteristics of FDI, despite the domestic investment. So, the relationship between FDI and economic growth may not be predictable. (Weinhold & Nair-Reichert, 2001). Kurtishi-Kastrati (2013) explained the FDI in the sense of microeconomics and international trade theories. Host country's

economic and political conditions usually positively affect the FDI inflow. Makiela & Ouattara (2018) investigated the effect of FDI on total factor productivity growth in two sample groups and concluded that there is no strong, statistically significant relationship between TFP and FDI.

Another research issue in literature is the FDI attracting countries such as Asian countries, especially China. For example, Weijun Wu, Ling Yuan, Xiaoming Wang, Xiaping Cao & Sili Zhou (2020) found a U-shaped relationship between FDI and economic growth for Chinese economy. FDI has an accelerating effect when its size is smaller than the local economy until a threshold, but when FDI crosses this point, its marginal effect decreases. It should be noted that with FDI, there is a technology transfer as well. Their impact on the economy has been analyzed in Lee & Tan (2006) that there are varied FDI intensities among four ASEAN countries; Singapore, Malaysia, Thailand and Indonesia. Singapore is the most competitive one. *Malaysia ranked as second followed by Thailand and lastly Indonesia in terms of FDI intensities and technology transfer intensities.* So, the technology transfer may be seen as an ability to absorb by economy and then to spillover. The South Korean economy is another miracle in Asian economies. It was a poor country in the 1960s but after then, *the massive inflow of FDI and double-digit growth in the economy in South Korea since 1980* is worth studying. (Koojaroenprasit, S, 2012) A good example of FDI and economic growth relationship.

FDI is seen as a significant contributor of economic growth in OECD countries for many studies. Bhattarai (2016) studied the macro and micro foundations of FDI and its impact on 30 OECD countries. He showed positive effects of FDI inflows and negative impacts of FDI outflows on economic growth and investment. However, in some studies it is found that FDI has no big contributions on accelerating growth in developed economies but it is *moderate*. (Ghosh & Wang, 2010). Although traditionally it is widely accepted that FDI inflow engines economic growth, the opposite direction of causation is valid too. Türkcan & Duman & Yetkiner (2020) studied the endogenous relationship between foreign direct investment and economic growth. The study shows FDI positively affects economic growth rate and also economic growth rate positively affects FDI inflows, no surprisingly. *But economic growth stimulates growth rate of FDI inflows more than that the growth rate of FDI stimulates economic growth*, a different point of view from other studies before. Generally, the relationship is prevailing across OECD countries, for Turkey's case, there is found no relationship between foreign direct investments and economic growth

in Turkey's economy both in the short run and in the long-run. (Ünsal,2017). Also, I need to add that as the countries maintain the quality of their institutions, they will do better in economic growth and FDI inflows, and good governance. (Raza & Shah & Arif, 2019).

In the studies that examine interest rate and economic growth, there are various results different from each other in some aspects. In the study of OECD economies in 1960-2010, which is based only on 12 countries, the relationship between growth, inflation and interest rate is examined. It realizes the negative relationship between GDP growth and interest rate. Inflation did not have an effect on this relationship. (Salim Ergene, 2015)

Another study examines the relationship between interest rate and GDP growth with the data of Germany, Japan, the UK and the US. There is statistical causality which has shifted from growth to interest rate. And we can clearly see that growth affects interest rate but interest rate is not that much affected by growth. (Lee & Wernera, 2018)

Another economic study examines the relationship between interest rate and growth in Latvia, which had the highest economic growth rate of the EU, is based on the years of 1996-and 2019. Interest rate and GDP growth was negatively related to each other. And also, while real interest rate caused GDP growth in the short run, GDP growth caused real interest rate in the long run. (Goliuk V.Y, 2018)

Another study that examines the Japan 1981-2008 data with Hansen model and the model shows the real interest rate has an important threshold effect. The effect of the banking system on GDP growth is positive in the regime where the prevailing real interest rate level is higher. The effect of finance on growth is insignificant, in the regime with prevailing real interest rate levels low. (Chang & Huang, 2010)

As for the relationship between unemployment and economic growth, we examined several papers. They generally refer to the Okun's law. The law, which Okun presented in his article in 1962, states the relationship between the GDP and the unemployment rate as negative. Mucuk et al. (2017) used Johansen cointegration test and vector error correction model for the case of Turkey and empirical findings did not show up a result supporting Okun's law. Tatoğlu (2011) examined the subject relation over the 1977-1988 period across European countries using panel unit root, panel cointegration and panel error correction model. Tatoğlu concluded that the validity of Okun's law and the relationship between the unemployment rate and the GDP vary between countries. Soyulu et al (2018) have analyzed the Eastern European

countries for 1992-2014 using panel data, panel unit root tests, pooled OLS estimation and panel cointegration test. They have concluded on; the unemployment rate would fall by 0.08% with 1% rise in GDP. Harris and Silverstone (2001) have tested the Okun's law's assumption of symmetric relationship between unemployment and changes in GDP for seven OECD countries, and their work showed no empirical evidence for the relationship in the long-run. Lee (2001) has analyzed Okun's law's robustness by using postwar data of 16 OECD countries and the dataset generally supported the validity of Okun's law. He concluded on cointegration between unemployment and growth.

3. Data Review

This study works on the countries of the Organization for Economic Co-operation and Development (OECD). If we give a brief description about this organization, which was established to support and restore Western European economies after World War II. It has many tasks such as providing statistical, economic and social data, analyzing economic developments and making predictions. OECD countries are high-income economies which are known as developed countries. According to the World Economic Outlook Database in April 2018, OECD member countries accounted for 62.2% of global nominal GDP in purchasing power parity from 2017.²

In this study, to explain the growth of Gross Domestic Product which is dependent variable, the independent variables are Consumer Confidence Index(CCI), Foreign Direct Investment(FDI), Short term Interest Rate (IR) and Unemployment Rate(UR) for the 37 OECD countries namely Australia, Austria, Belgium, Canada, Switzerland, Chile, Colombia, Czech Republic, Germany, Denmark, Spain, Estonia, Finland, France, United Kingdom, Greece, Hungary, Ireland, Iceland, Israel, Italy, Japan, Korea Republic, Lithuania, Luxembourg, Latvia, Mexico, Netherlands, Norway, New Zealand, Poland, Portugal, Slovak Republic, Slovenia, Sweden, Turkey and United States. The data for the dependent and independent variables is taken from OECD website but only data of Foreign Direct Investment are taken from World bank website. This study is based on 2019. Main purpose of this study is to explain the relationship between GDP Growth and Consumer Confidence Index in the short run.

² World Economic Outlook Database". International Monetary Fund. 17 April 2018

The definitions of the variables are as follows:

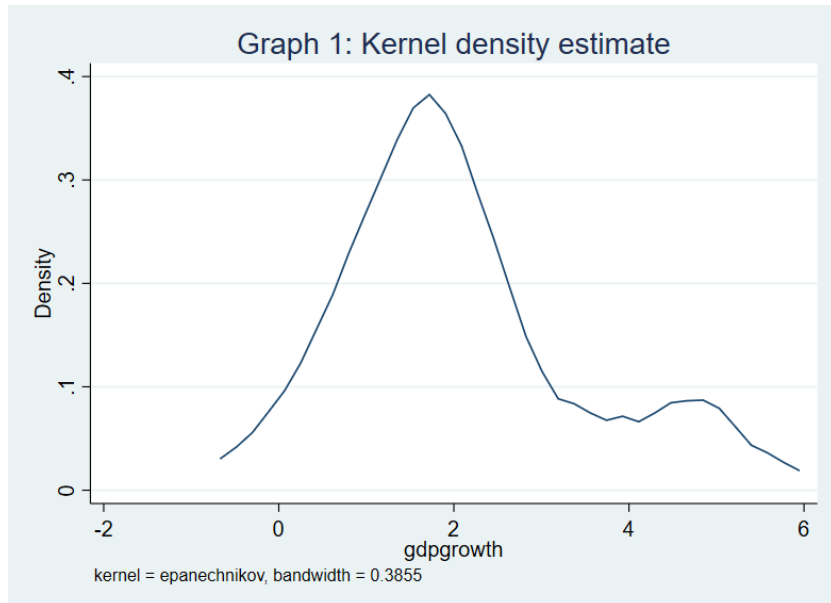
- GDP Growth: GDP, volume – annual growth rates in percentage. GDP is the sum of gross value added by all resident producers in the economy and also product taxes and subsidies not included.
- Consumer Confidence Index (CCI): is an indicator that aims to measure the expenditure and saving trends in the near future, based on the current situation evaluations of the consumers' financial conditions and the general economy and their future expectations. According to OECD, who are taken 100 as an indicator and divide our observations into CCI below 100 and those above 100. In the first group, those below 100, which have poor expectations for the future economy and tend to save more and consume less. In the second group, those above 100, which have positive expectations for future economic situations, and they are less likely to save and tend to spend more money.
- Foreign Direct Investment (FDI): is the value of cross-border transactions related to direct investment during a year. FDI flows are measured in USD and as a share of GDP.
- Short Run Interest Rate (IR): is averages of daily rates and measured with percentage. It affects short-term borrowings between financial institutions and government paper in the market.
- Unemployment Rate (UR): has some rules. Firstly, people of working age who are without work and available for work. Secondly, they look for a job. Unemployment rate measured in numbers of unemployed people as a percentage of the labor force.

We used abbreviations for the mentioned variables on Stata. From now on, the abbreviations will also be used from time to time. They are respectively “gdpgrowth, cci, fdi, interstrate and unemployment” corresponding “GDP growth, Consumer Confidence Index, Foreign Direct Investment, Short Run Interest Rate and the Unemployment Rate”.

Table 1: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
gdpgrowth	37	2.068	1.388	-.285	5.565
cci	34	100.82	1.684	96.362	103.988
fdi	37	23022.011	47773.903	-21740.883	282053
interestrate	36	.765	1.875	-.738	8.273
unemployment	37	5.794	3.326	1.933	17.238

After the definition of the variables, we are ready to look at summary statistics with Table 1. As we mentioned before, we want to study in 37 countries which are OECD member countries. But missing value problems have occurred in the data for some countries which are for CCI data in Canada, Iceland, Norway and Interest Rate data in Turkey. Therefore, we had to narrow our observations. And also, we clearly see that the variables are in expected range. The mean of the GDP growth is 2.068 and it shows that all countries in the study grew %2.068 on average. The rate of Ireland, which has the highest growth rate, is 5%. In addition, the mean of CCI is 100,82 which means that all countries have positive expectations for the future economic situation, meaning their demand and spending would increase. GDP growth is also indirectly affected by CCI.



After the summary statistics, examine the distribution of our dependent variable with Graph 1. The distribution of GDP growth is close to the normal distribution but there seems to be a strangeness. It has a bimodal distribution. GDP growth of countries is separated into two parts with two modes. Right mode has the higher GDP growth countries which are Ireland, Estonia, Hungary, Poland, Lithuania. If we look at the OECD countries' general situation, we clearly see that they have high GDP growth.

4. Model

The purpose of this study is to explain the short run relationship between GDP growth and Consumer Confidence Index with the other independent parameters which are mentioned in the Data set part. We estimate the model with Ordinary Least Squares. We choose it because we are studying with cross section data. And also, this method has an advantage which serves to find a function curve that will pass in as close as possible to the data points obtained from the measurement result. Although this model explains the GDP growth with a rate of 30 percent, this rate is a good rate for an economic model. In the data set section, we have made the definitions of the variables, here we symbolized the model.

$$\text{GDP}_{\text{GROWTH}} = \beta_0 + \beta_1 \text{CCI} + \beta_2 \text{FDI} + \beta_3 \text{IR} + \beta_4 \text{UR} + \epsilon$$

Where, growth of Gross Domestic Product, which is dependent variable, the independent variables are Consumer Confidence Index (CCI), Foreign Direct Investment (FDI), Short term Interest Rate (IR) and Unemployment Rate (UR) and ϵ denotes the error terms, respectively.

The coefficient of the term CCI (β_1) shows that there is a linear relationship between economic growth and CCI.

We evaluate the robustness of the model with some tests. First, we check the model for heteroskedasticity problem and the model has not included which means the residuals are homoscedastic.

Secondly, we then prepared a scatterplot of the fitted values versus residuals for our regression. We observed 2 outliers which are

greater than 2, as can be seen on the Graph 2, are worth investigating. Residual countries for the model are Colombia and Ireland.

Thirdly, we test using powers of the fitted values of gdp_{growth} the model which checks the omitted variable and not include omitted variable.

We also made the IM-test to provide an overview of the regression model and it shows the information matrix test for the regression model and an orthogonal decomposition into tests for heteroskedasticity, skewness, and kurtosis (can be seen at Table 2 below). There is no heteroscedasticity in our model. As we mentioned in the data set section, kurtosis and skewness are very close to normal distribution.

End of the tests, we observe that our model is suitable for working on.

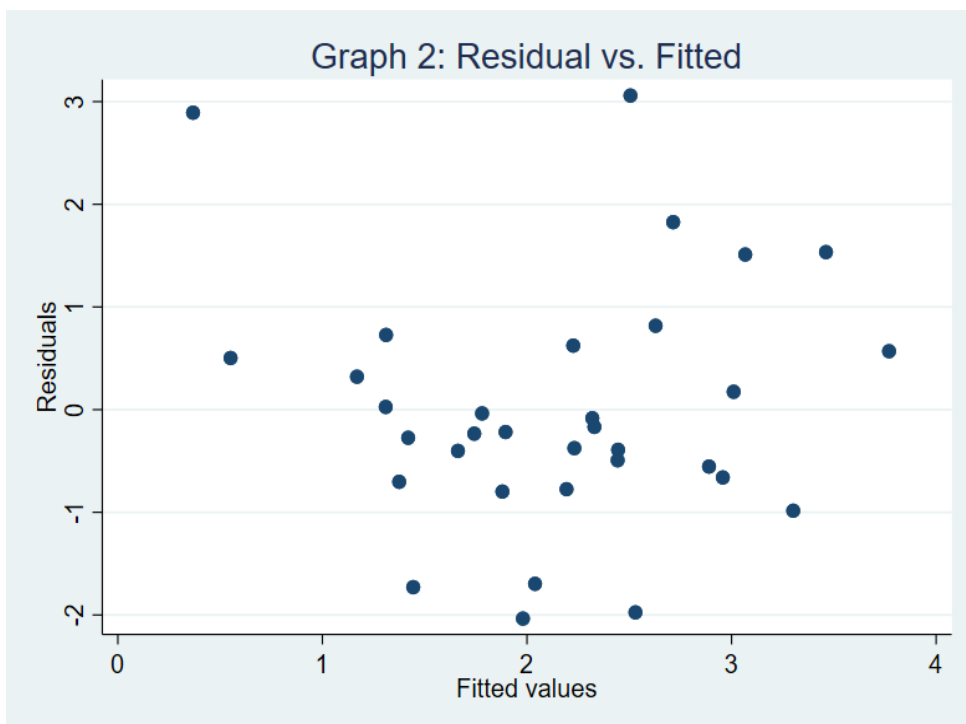


Table 2:
Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
Heteroskedasticity	26.620	14	0.021
Skewness	5.880	4	0.208
Kurtosis	1.020	1	0.312
Total	33.530	19	0.021

5. Empirical Results

Table 3: Matrix of correlations

Variables	(1)	(2)	(3)	(4)	(5)
(1) gdpgrowth	1.000				
(2) cci	0.491	1.000			
(3) fdi	-0.015	-0.024	1.000		
(4) interestrates	-0.192	0.088	0.206	1.000	
(5) unemployment	-0.073	-0.114	-0.156	-0.153	1.000

First, we found the correlation coefficients for the variables, as can be seen on Table 3, correlation between the gdpgrowth and the cci is high as it is expected. This explains why the countries with high CCI levels have high GDP growth levels. Other variables are not really correlated with the dependent variable. Also, independent variables have low correlation between them, which means we do not have to worry about a multicollinearity problem.

We then prepared a graph between gdpgrowth and cci with a scatter plot and a regression line, can be seen at Graph 3. And we also ran a regression between gdpgrowth and cci only, and the result is $\text{gdpgrowth} = -40.6 + 0.42 * (\text{cci})$. The probability values for cci and constant are respectively 0.003 and 0.004 making them significant at %1 level. The r-squared for the mentioned regression is 0.2485, meaning the cci only is not explaining the variation much. R-squared value increases, but not that much, as we include other variables in the model as it can be seen below at our main regression.

On the Graph 3, Mexico attracts attention considering the CCI level (which is one of the highest) and the GDP growth level (which is one of the lowest). All other countries do not really disaccord with our pre-mentioned expectations.

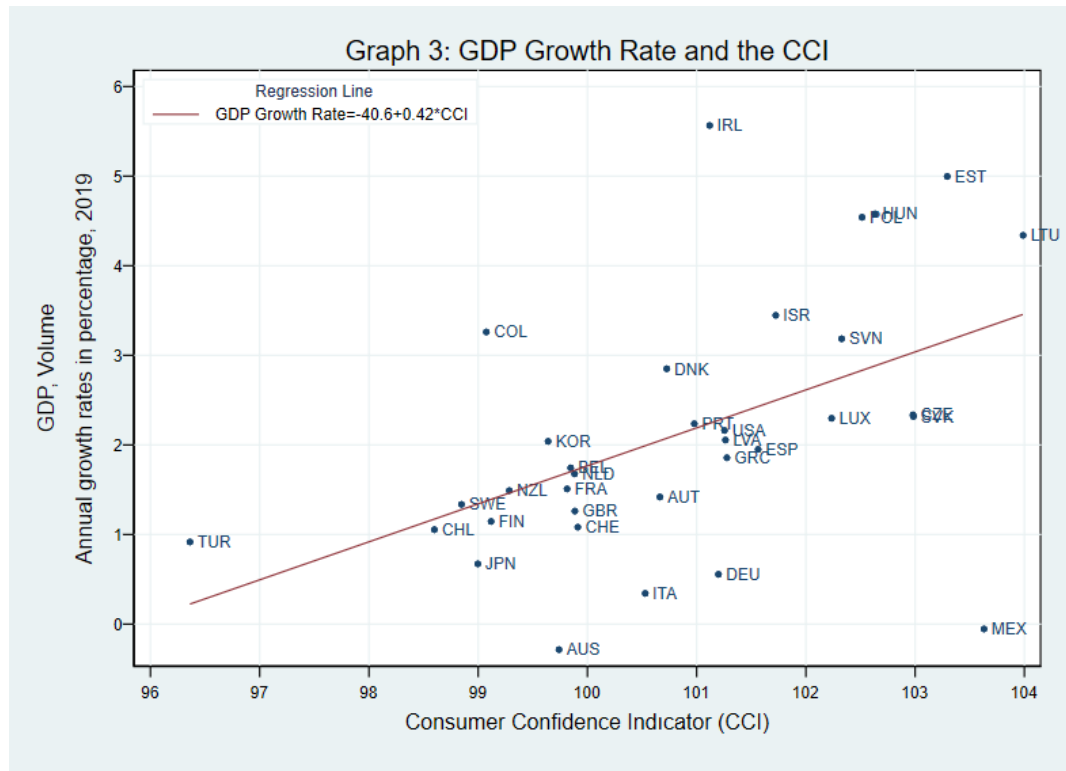


Table 4: Linear Regression:

Gdpgrowth	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Cci	.485	.148	3.27	.003	.181	.789	***
Fdi	0	0	0.37	.717	0	0	
Interestrates	-.199	.221	-0.90	.376	-.652	.254	
Unemployment	-.021	.061	-0.34	.735	-.147	.105	
Constant	-46.567	14.775	-3.15	.004	-76.832	-16.301	***
Mean dependent var		2.150	SD dependent var		1.439		
R-squared		0.302	Number of obs		33.000		
F-test		3.595	Prob > F		0.017		
Akaike crit. (AIC)		114.798	Bayesian crit. (BIC)		122.280		

*** $p < .01$, ** $p < .05$, * $p < .1$

We ran an OLS regression with heteroskedasticity-robust standard errors. The overall probability value 0.017 means that regressors are jointly statistically significant at 0.01 level. R-squared is 0.302, the variation is not that much explained. "cci" has the p-value of 0.003 and constant has the p-value of 0.004 means that both of them are statistically significant. Other variables are not statistically significant.

We expected cci to have a positive impact on gdpgrowth and the coefficient .485 confirms us. We expected fdi to have an impact on gdpgrowth but within our study it is statistically insignificant, it has a coefficient of 0. Interest rate here on our study results has a negative relationship on gdpgrowth as expected but unfortunately it has no significant impact. Again, unemployment also has negative relationship but there is no significance.

Our study shows, for the year and country group obviously, as the only important variable is CCI. And if we consider the policy-making with what we have got, preserving only the consumer confidence would serve best to the growth-seeking attitude. The rest of the economic indicators could be ignored, which is very unreal. Consequently, our regression is not sufficient enough to explain the gdpgrowth by all variables, which is why we have a low r-squared value.

Now we can conclude on our primary question which is “Does CCI have an impact on economic growth?”. Yes, it does and it has a significant impact within selected countries by year 2019.

We applied Breusch-Pagan / Cook-Weisberg test for heteroskedasticity and we got:

Ho: Constant variance, fitted values of `gdpgrowth` as variables, $\chi^2(1) = 0.33$ and $\text{Prob} > \chi^2 = 0.5658$. We cannot reject the null hypothesis, that the residuals are homoscedastic.

We also applied the Ramsey reset test using powers of the fitted values of `gdpgrowth` and we got:

Ho: model has no omitted variables. $F(3, 25) = 4.02$ and $\text{Prob} > F = 0.0183$. At the 0.01 significance level, we fail to reject the null hypothesis. There are no omitted variables.

Table 5: Link Test

Source	SS	df	MS	Number of obs	=	33
Model	29.1701093	2	14.5850546	F(2, 30)	=	11.81
Residual	37.064757	30	1.2354919	Prob > F	=	0.0002
				R-squared	=	0.4404
				Adj R-squared	=	0.4031
Total	66.2348662	32	2.06983957	Root MSE	=	1.1115

<code>gdpgrowth</code>	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
<code>_hat</code>	-1.766492	1.044028	-1.69	0.101	-3.898681	.3656974
<code>_hatsq</code>	.6629363	.2429784	2.73	0.011	.1667082	1.159165
<code>_cons</code>	2.482284	1.072918	2.31	0.028	.2910929	4.673476

We decided to run a postestimation test, namely link test, that creates two new variables which are: the variable of prediction (`_hat` on Table 5) and the variable of squared prediction (`_hatsq` on Table 5). It is being predicted value, `_hat` should be significant. And `_hatsq` should not have explanatory power, being squared prediction we would expect it to be insignificant. P-value of `_hatsq` is 0.011, meaning that it is insignificant at 0.01 level. A significant `_hatsq` would indicate an

omitted variable problem and with our χ^2 is insignificant, we could say that there is no omitted variable problem at 0.01 level which is consistent with our finding on Ramsey reset test.

6. Conclusion

This paper argues the consumer confidence index (CCI) component of economic growth in OECD countries for 2019.

According to the OECD and World Bank data, GDP growth has not a similar pattern among countries. We investigated the further effects of consumers' attitude on the economy via consumer confidence index (CCI). Especially in Nordic countries, Canada, Iceland and Norway which we expected high levels of CCI in these countries as well as their high GDP levels, there is not available CCI data for 2019. Also, there is not a universal consumer confidence measure so some countries use different indexes for consumer confidence.

According to the previous studies in literature, there is a strong linkage between CCI and economic growth. We have questioned this assessment through OECD countries. It is found that countries with higher CCI levels have higher economic growth, respectively. Thus, CCI has a positive impact on (GDP growth) economic growth that shares common sense in the literature.

However, Mexico is an interesting country that does not share the same pattern among OECD countries. Mexico has high CCI however its GDP growth is not high, contrary to expectations. It seems that higher CCI has no impact on GDP growth of Mexico in 2019.

Economic growth is a complex issue as well as CCI data. Consumers' expectations have significant influences on behavior and that determines the market's conditions in the economy. It can be concluded that it should be more studied in CCI and economic growth in some countries, especially in Mexico according to the results of this paper. It might be a special case for 2019 year as well. It is worth investigating more. If we can learn more about this relationship effectively, new studies may offer a remarkable new tool for policymakers.

References

1. Arrazy, M. R. (2013). Factors That Influence GDP Growth Rate in Six Asia's Developing Countries From 2007 To 2011 (Doctoral dissertation, President University).
2. Bhattarai, K. (2016). FDI and growth. *Advances in Management and Applied Economics*, 6(2), 1-23
3. Chang, S. H., & Huang, L. C. (2010). The nexus of finance and GDP growth in Japan: Do real interest rates matter?. *Japan and the World Economy*, 22(4), 235-242.
4. Dees, S., & Brinca, P. S. (2013). Consumer confidence as a predictor of consumption spending: Evidence for the United States and the Euro area. *International Economics*, 134, 1-14.
5. Dominitz, J., & Manski, C. F. (2004). How should we measure consumer confidence?. *Journal of Economic Perspectives*, 18(2), 51-66.
6. Eldin Mehic , Sabina Silajdzic & Vesna Babic-Hodovic (2013) The Impact of FDI on Economic Growth: Some Evidence from Southeast Europe, *Emerging Markets Finance and Trade*, 49:sup1, 5-20, DOI: 10.2753/REE1540-496X4901S101
7. Ergene, S. (2015). Growth, inflation, interest rate and informality: Panel VAR evidence from OECD Economies. *Economics Bulletin*, 35(1), 750-763.
8. Ghosh, Madanmohan & Wang, Weimin. (2010). Does FDI Accelerate Economic Growth? The OECD Experience Based on Panel Data Estimates for the Period 1980-2004. *Global Economy Journal*. 9. 1-1. 10.2202/1524-5861.1496.
9. Goliuk, V. Y. (2018). The effect of interest rate policy on GDP dynamics in Latvia.
10. Harris, R., & Silverstone, B. (2001). Testing for asymmetry in Okun's law: A cross-country comparison. *Economics bulletin*, 5(2), 1-13.
11. Huay Huay Lee & Hui Boon Tan (2006) Technology Transfer, FDI and Economic Growth in the ASEAN Region, *Journal of the Asia Pacific Economy*, 11:4, 394-410, DOI: 10.1080/13547860600923593
12. Islam, T. U., & Mumtaz, M. N. (2016). Consumer Confidence Index and Economic Growth: An Empirical Analysis of EU Countries. *EuroEconomica*, 35(2).
13. Jalilian, H., Kirkpatrick, C., & Parker, D. (2007). The impact of regulation on economic growth in developing countries: A cross-country analysis. *World development*, 35(1), 87-103.
14. Kamil Makiela, Bazoumana Ouattara, Foreign direct investment and economic growth: Exploring the transmission channels, *Economic Modelling*, Volume 72, 2018, Pages 296-305, ISSN 0264-9993,
15. Koojaroenprasit, S. (2012). The impact of foreign direct investment on economic growth: A case study of South Korea. *International Journal of Business and Social Science*, 3(21).
16. Kurtishi-Kastrati, S. (2013). Impact of FDI on economic growth: An overview of the main theories of FDI and empirical research. *European Scientific Journal*, 9(7).
17. Lee, J. (2000). The robustness of Okun's law: Evidence from OECD countries. *Journal of macroeconomics*, 22(2), 331-356.

18. Lee, K. S., & Werner, R. A. (2018). Reconsidering monetary policy: An empirical examination of the relationship between interest rates and nominal GDP growth in the US, UK, Germany and Japan. *Ecological Economics*, 146, 26-34.
19. Mazurek, J., & Mielcová, E. (2017). Is consumer confidence index a suitable predictor of future economic growth? Evidence from the USA. *Economics and Management*.
20. Moudatsou, A., & Kyrkilis, D. (2011). FDI and Economic Growth: Causality for the EU and ASEAN. *Journal of Economic Integration*, 26(3), 554-577. Retrieved December 24, 2020
21. Mucuk, M., Edirneligil, A., & Gerçeker, M. (2017). The relationship between unemployment rate and economic growth: The case of Turkey. *Siyaset, Ekonomi ve Yönetim Araştırmaları Dergisi*, 5(1).
22. Panagiotis Pegkas, The impact of FDI on economic growth in Eurozone countries, *The Journal of Economic Asymmetries*, Volume 12, Issue 2, 2015, Pages 124-132, ISSN 1703-4949
23. Popescu, G.H. FDI and Economic Growth in Central and Eastern Europe. *Sustainability* 2014, 6, 8149-8163.
24. Raza, Syed & Shah, Nida & Arif, Imtiaz. (2019). Relationship Between FDI and Economic Growth in the Presence of Good Governance System: Evidence from OECD Countries. *Global Business Review*. 097215091983348
25. Sergeant, K., Lugay, B., & Dookie, M. (2011). Consumer confidence and economic growth: A case study of Trinidad and Tobago. In 4th International Conference on Business, Banking & Finance.
26. Sorić, P. (2018). Consumer confidence as a GDP determinant in New EU Member States: a view from a time-varying perspective. *Empirica*, 45(2), 261-282.
27. Soylu, Ö. B., Çakmak, İ., & Okur, F. (2018). Economic growth and unemployment issue: Panel data analysis in Eastern European Countries.
28. Tatoğlu, F. Y. (2011). The long and short run effects between unemployment and economic growth in Europe.
29. Türkcan, Burcu & Duman, Alper & Yetkiner, I.. (2020). How Does FDI and Economic Growth Affect Each Other? The OECD Case.
30. Ünsal, M. E. (2017). FDI and Economic Growth: Comparative Analyses between Turkey and the other OECD Countries. *Journal of Current Researches on Business and Economics*, 7(2), 207-2016.
31. Weijun Wu, Ling Yuan, Xiaoming Wang, Xiaping Cao & Sili Zhou (2020) Does FDI Drive Economic Growth? Evidence from City Data in China, *Emerging Markets Finance and Trade*, 56:11, 2594-2607, DOI: 10.1080/1540496X.2019.1644621
32. Weinhold, Diana & Nair-Reichert, Usha. (2001). Causality Tests for Cross-Country Panels: A New Look at FDI and Economic Growth in Developing Countries. *Oxford Bulletin of Economics and Statistics*. 63. 153-71. 10.1111/1468-0084.00214.